NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-6106/20B(USAF)
24 October 1990

SUPERSEDING
MIL-R-6106/20A(USAF)
7 January 1985

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, MAGNETIC LATCH, 25 AMPERES, SPDT, HERMETICALLY SEALED

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification list in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-PRF-6106.

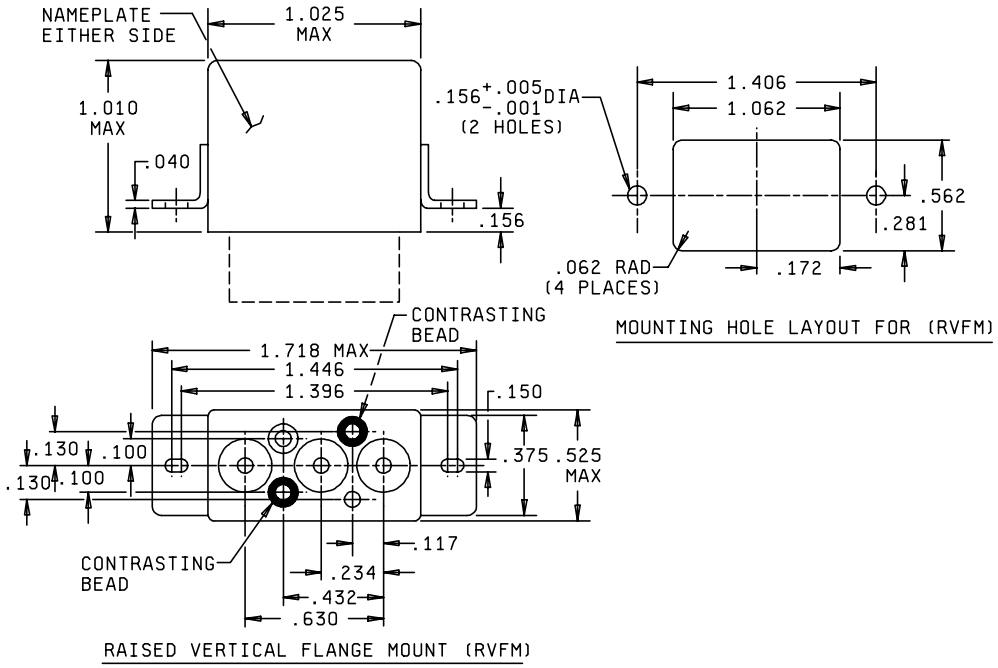
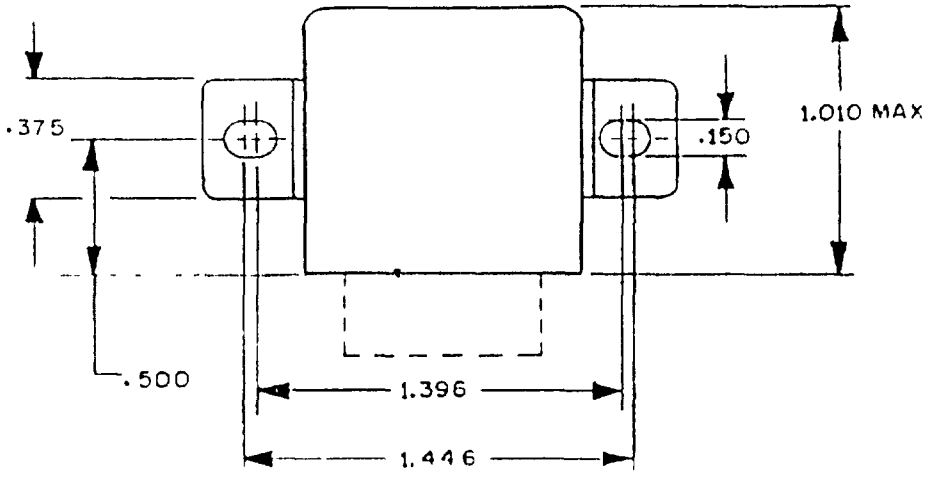
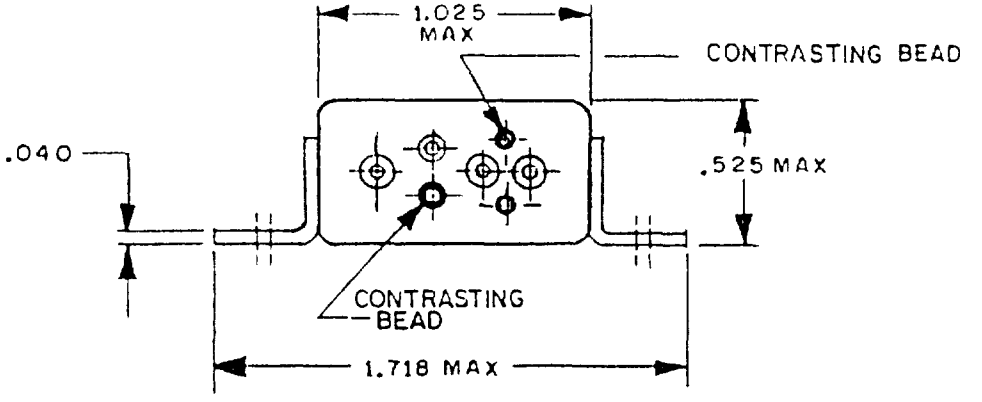


FIGURE 1. Relay, outline drawing.

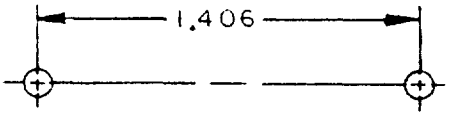
DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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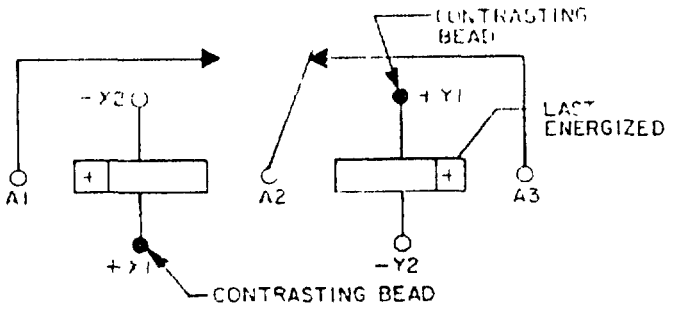
(B) HORIZONTAL FLANGE MOUNT (HFM)



(B) MOUNTING HOLE LAYOUT FOR HFM

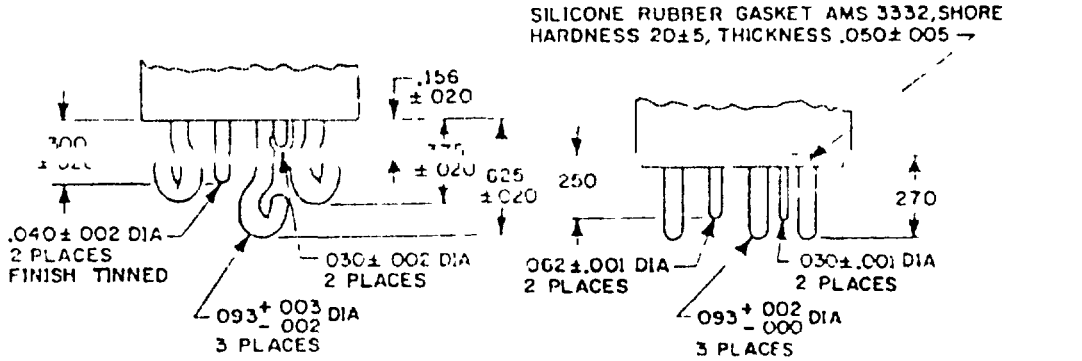
FIGURE 1 Relay, outline drawing - Continued

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CIRCUIT DIAGRAM
(POLARIZED BISTABLE)

- TO CLOSE NO 1 CONTACTS (OPERATE) ENERGIZE X1 AND X2
- TO CLOSE NO 3 CONTACTS (RESET) ENERGIZE Y1 AND Y2



SOLDER HOOK TERMINAL FOR -001 AND -003

FINISH GOLD PLATE, MIL-G-45204, TYPE II, CLASS 1 UNDERPLATING, NICKEL 50 TO 150 MICROINCHES THICK

PLUG-IN TERMINAL FOR -002

Inches	mm	Inches	mm	Inches	mm
001	0.03	.130	3.30	.525	13.34
.002	0.05	150	3.81	.562	14.27
003	0 08	156	3 96	.625	15.88
.005	0 13	172	4 37	.630	16 00
.020	0 51	.234	5 94	1 010	25 65
030	0.65	.250	6 35	1.025	26 04
040	1.02	.270	6 86	1 062	26 97
050	1.27	281	7 14	1.396	35.46
.062	1.57	.300	7.62	1.406	35.71
093	2 36	375	9 53	1 446	36 73
100	2 54	432	10 97	1 718	43.64
117	2 96	500	12.70		

NOTES

1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ± 010 (25 mm)
4. Terminal numbers need not appear on the relay header provided there is affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified
5. Note to observe polarity must appear on relays
6. Hook terminals will accept one No 12 AWG stranded wire

FIGURE 1 Relay, outline drawing - Continued

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REQUIREMENTS:

Weight 0.10 pound maximum

Temperature range 1/ -70°C to +125°C

Maximum altitude-rating 80,000 feet

Shock G-level 200 G

Duration 6 milliseconds

Maximum duration contact opening 10 microseconds

Vibration-sinusoidal

G-level 30 G

Frequency range 10-3,000 Hz

Vibration-random

(U) Applicable specification MIL-STD-202, method 214, test condition 16

Duration 15 minutes, each plane

Acceleration 15 G

Insulation resistance, initial 100 megohms

After life or environmental tests 50 megohms.

Dielectric withstanding voltage (sea level)

	Initial	After life tests
Coil to case and coil to coil	1,000 V rms	1,000 V rms
All other points	1,000 V rms	1,000 V rms

Dielectric withstanding voltage (altitude) 2/

	80,000 feet	300,000 feet
Coil to case and coil to coil	250 V rms	500 V rms
All other points	350 V rms	500 V rms

1/ For full rated load, maximum temperature, and altitude, use no. 12 wire or larger. Solder hook relays will be mounted to limit mounting bracket temperature to 135°C.

2/ Low altitude dielectric withstanding voltage rating may be improved by suitable insulation of terminals and wiring after installation. Plug in relays, with suitable gasket compressed, will have dielectric withstanding voltage capabilities at 80,000 feet over 500 V rms.

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Maximum contact drop, initial 0.150 volt.

After life test 0.175 volt

Overload current 50 amperes dc, 80 amperes ac, 400 Hz.

Rupture current 60 amperes dc, 100 amperes ac, 400 Hz.

Duty rating Continuous

Ⓑ Part or Identifying Number (PIN). M6106/20- (plus dash number from table I) Example: M6106/20-001.

Ⓑ TABLE I. Dash numbers and operating characteristics

	Coil data								Time-milliseconds maximum 1/					
		Nominal		Max		Max pick-up voltage					Bounce			
Dash numbers	Coil	Volts (V dc) 2/	Res Ω ±10%	Volts	Amp	Nominal 3/	High temp test	Cont current test	Operate	Reset	Main NO	Main NC	Terminal type	Mounting style
-001	X1-X2 Y1-Y2	28	600	29	050	18	19.8	22.5	10	10	1.0	1.0	Solder hook	RVFM
-002	X1-X2 Y1-Y2	28	600	29	050	18	19.8	22.5	10	10	1.0	1.0	Plug-in	RVFM
-003	X1-X2 Y1-Y2	28	600	29	050	18	19.8	22.5	10	10	1.0	1.0	Solder hook	HFM

1/ With nominal coil voltage.

2/ Caution: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay

3/ Over temperature range.

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TABLE II. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	50	25	25			25								2/
Inductive	10	12	12											
Inductive	20					15								
Motor	50	10	10			10								
Lamp	50	5	5			5								
Transfer, load														3/
Mechanical life reduced current	200	6	6			6								
Intermediate current		APPLICABLE (See MIL-R-6106)												

1/ Absence of value indicates relay is not rated for 3 phase application.

2/ For full rated load maximum temperature and altitude use no. 12 wire or larger. Solder hook relays to be mounted to limit mounting bracket temperature to 135°C

3/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated

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Ⓑ Qualification by similarity. If the relays produced in accordance with this specification sheet are similar in design and construction to those covered by MIL-R-6106/19 (except for the coils, which shall be similar to MS27745), then reduced qualification by similarity is authorized. Manufacturers shall be qualified to MIL-R-6106/19 and MS27745. Qualification by similarity requirements are listed in table III.

Ⓑ TABLE III. Qualification by similarity

Inspection	Number of units
Group A acceptance tests	7 1/
Vibration 2/ Resistive load, ac Group A acceptance tests	2
Shock 2/ Inductive load, dc Group A acceptance tests	2
Acceleration 2/ Intermediate current 3/ Group A acceptance test	3

1/ Four units of dash number -002 and three units of dash number-003. All seven relays shall receive group A acceptance testing in accordance with MIL-R-6106. The seven relays shall then be subdivided into three groups as shown above for the remaining inspections.

2/ One unit of dash number -002 and one unit of dash number -003.

3/ Three units required for intermediate current testing.

CONCLUDING MATERIAL

Custodian
Air Force - 85

Review activity
Air Force - 80

Preparing activity.
Air Force - 85

Agent
DLA - ES

(Project 5945-F697-01)

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